

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CL/EC/ME/ CE/EE/IT) SEM - I Examination, 2009-10

MA - 101 Engineering Mathematics 1

Date: 14.12.09, Monday

Time: 2:00 p. m to 5:00 p. m.

Maximum Marks:70

Instructions:

1. Attempt all the questions in each section.
2. Assume suitable additional data if required.
3. Figures to the right side indicate the full marks.

SECTION-I

Q-1 If $p = \operatorname{cis} \theta$ and $q = \operatorname{cis} \phi$, show that (i) $\frac{p-q}{p+q} = i \tan \frac{\theta-\phi}{2}$ (ii) $\frac{(p+q)(pq-1)}{(p-q)(pq+1)} = \frac{\sin \theta + \sin \phi}{\sin \theta - \sin \phi}$ [06]

Q-2 Test the convergence of the following series: [09]

(i) $\frac{1}{1+2} + \frac{2}{1+2^2} + \frac{3}{1+2^3} + \dots$ (ii) $\sum_{n=1}^{\infty} \frac{\cos n\pi}{n^2+1}$ (iii) $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{1}{n^2+n+1} \right)$

OR

Q-2 Test the convergence of the following series: [09]

(i) $\sum_{n=1}^{\infty} \frac{(n+1)^n x^n}{n^{n+1}}$ (ii) $\frac{1}{1 \cdot 2 \cdot 3} + \frac{x}{4 \cdot 5 \cdot 6} + \frac{x^2}{7 \cdot 8 \cdot 9} + \dots$ (iii) $\sum_{n=1}^{\infty} \frac{3+2 \sin n}{n^3}$

Q-3 (i) Determine the rank and nullity of the matrix $\begin{bmatrix} 12 & 32 \\ 23 & 51 \\ 13 & 45 \end{bmatrix}$ [05]

(ii) Using Gauss-Jordan method, Find the inverse of $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ [05]

(iii) Test for consistency and solve the following system of equations: [05]

$$x + y + z = 6$$

$$x - y + 2z = 5$$

$$3x + y + z = 8$$

$$2x - 2y + 3z = 7$$

(iv) Find all the roots of $(1+i)^{\frac{1}{3}}$ and show that their continued product is $1+i$. [05]

OR

Q-3 (i) If $i^{a+ib} = \alpha + i\beta$, Prove that (i) $\alpha^2 + \beta^2 = e^{-\pi b}$ (ii) $\beta = \alpha \tan\left(\frac{\pi a}{2}\right)$ [05]

(ii) Expand $\sin^4 \theta \cos^2 \theta$ in a series of multiples of θ . [05]

(iii) Investigate for what values of K, simultaneous equations [05]

$$2x - 3y + 6z - 5t = 3$$

$$y - 4z + t = 1 \quad \text{have (a) no solution (b) infinite number of solutions.}$$

$$4x - 5y + 8z - 9t = k$$

(iv) Determine the rank and nullity of the matrix

$$\begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 \end{bmatrix}$$

SECTION-II

Q-4 (i) Find the n^{th} derivative of $y = \frac{x+1}{x^2-4}$ [03]

(ii) Evaluate: $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$ [02]

Q-5 (i) Express $\cos(a+h)$ as a series of powers of h and hence evaluate the value of $\cos 46^\circ$ correct up to three decimal places. [05]

(ii) State Roll's Theorem. Show that $x^3 + 3x^2 + 6x + 1 = 0$ has exactly one real root. [05]

(iii) Use Lagrange's Mean Value Theorem to prove [05]

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}, \text{ if } 0 < a < b.$$

$$\text{Hence deduce that } \frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{16}$$

OR

Q-5 (i) Expand $f(x) = \sec x$ in powers of x up to x^4 by Maclaurin's series. [05]

(ii) State Cauchy's Mean Value Theorem. Verify it for functions $f(x) = e^x$ and $g(x) = e^{-x}$ defined on (a, b) and show that 'C' of the Cauchy's Mean value theorem is the average of 'a' and 'b'. [05]

(iii) Use Lagrange's Mean Value Theorem to prove $1 - \frac{a}{b} < \ln \frac{b}{a} < \frac{b}{a} - 1$, and hence [05]

$$\text{Show that } \frac{1}{6} < \ln \frac{6}{5} < \frac{1}{5}.$$

Q-6 (i) State Euler's Theorem on Homogeneous functions. [05]

If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, Show that

(a) $x u_x + y u_y = \sin 2u$
(b) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 2 \sin u \cos 3u$

(ii) A rectangular box open at the top is to have a volume of 32 cubic feet. Find the dimensions of the box requiring minimum material for its construction. [05]

(iii) Prove that $x^3 y^2 (12 - x - y)$ is maximum at the point (6, 4). Find its maximum value. [05]

OR

Q-6 (i) Find the percentage error in computing the resistance r from the formula [05]

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}, \text{ if } r_1, r_2 \text{ are both in error by } 2\%.$$

(ii) Using Lagrange's method, Obtain the values of x, y, z which will minimize [05]

$$x + y + z \text{ subject to the condition } \frac{4}{x} + \frac{9}{y} + \frac{16}{z} = 25.$$

(iii) If $u = x + y^2, v = y + z^2, w = z + x^2$, find $J = \frac{\partial(u, v, w)}{\partial(x, y, z)}$ and $J' = \frac{\partial(x, y, z)}{\partial(u, v, w)}$. Show that [05]

$$JJ' = 1.$$

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